

Do Regime-Aware Monte Carlo Methods Improve VaR and CVaR Forecasting? Evidence from Eight Global Equity Markets

Vishwa Jadhav

Department of Computer Science and Engineering

Amity University Mumbai, Maharashtra, India

vishwa.jadhav@s.amity.edu

Abstract

Both volatility filtering and regime switching are strongly suggested for forecasting extreme events in tails but comparison between the two is absent, and not always done in a standardized and systematic fashion across the market using a common out-of-sample test. The paper compares the performance of five one-step-ahead Value-at-Risk (VaR) models and Conditional VaR (CVaR, equivalently expected shortfall, ES) models in terms of their effectiveness on close to 20 years of daily data obtained for 4 developed equity indices — S&P 500, DAX, FTSE 100, Nikkei 225 — and 4 emerging equity indices — Nifty 50, Sensex, Bovespa, and JSE Top 40. They are evaluated under Kupiec unconditional coverage, Christoffersen conditional coverage, McNeil–Frey expected-shortfall loss, and pairwise Diebold–Mariano (DM) tests, all using an identical rolling walk forward design of evaluation. The model whose calibration does not change according to the regime chosen — regime-aware Monte Carlo — is the only model to be fully calibrated on the expected shortfall for all 8 markets (16/16 cells) and the only one to be well calibrated in terms of 5% conditional coverage for all 8 markets (8/8). The HMM-Regime model is the worst on the 5% conditional coverage (0/8) and the worst on the fully calibrated expected shortfall (16/16 cells), with regime-aware Monte Carlo being clearly the winner on 32 out of its 57 comparisons (more than four times as many) in terms of predictive accuracy. Even the winner has a secondary result: at the 5% tail FHS-GARCH is still well calibrated across all markets, albeit with occasional breaches, the majority of which correspond to the crisis of 2008 or 2020, but calibration is only breached in the 1% tail from the developed markets. In the 16 cells, the Gaussian model under-sizes the tail, which is not what is shown by the expected-shortfall test. While the results indicate that while explicit discrete regime identification is important for practical tail-risk forecasting in broader equity indices, continuously modelling conditional volatility appears to be more important, and that the deepest-tail calibration of even the best approach is limited by clustering of crisis-time breaches in developed markets.

1. Introduction

The weakness of risk management is the problem that motivates this work. This work is motivated by a problem: The fragility of risk management.

The dynamic nature and volatility of today's equity markets always threaten the stability of the international financial system. In the aftermath of the 2008 financial crisis, quantitative risk evaluation has become a key strategy and the focus of the front-office business line for financial institutions. However, the most important problem for the risk managers is that financial markets have a “regime-dependent” characteristic, which means they tend to shift

between relatively calm and very turbulent periods. Normally these transitions are poorly captured by conventional risk models that make assumptions of normality or linear correlation which become less valid in times of market stress. The world markets are being increasingly linked; a shock in one market can easily spread to the others. Sensitive forecasting techniques are thus needed that can be accurate for various market states, so that institutions can have adequate capital buffers for catastrophic events and maximise their allocation during market growth.

To find out why VaR / CVaR, read on. Since it was introduced many decades ago, Value-at-Risk has been the industry standard for measuring market risk because of its very conceptual simplicity that captures the market risk in just one number. The drawbacks of VaR are far from trivial, however, the most obvious being the failure to be sub additive (the risk of a diversified portfolio can seem to be larger than the sum of its parts), and ignoring the "tail risk" beyond the quantile. Such shortfalls were bitterly felt in times of crisis when "once in a century" events happened too often.

To counter this, the Basel Committee on Banking Supervision launched a change in the paradigm with Fundamental Review of the Trading Book and Basel III. This regulatory shift introduced a new measure of market risk capital requirements that is called Expected Shortfall, or Conditional Value-at-Risk. CVaR is a "coherent" risk measure which measures the average loss in the tail of the distribution, unlike VaR which does not. This shift in regulation is a shift in regulatory thinking, in that it is one based on the severity of the event instead of simply the frequency of the event, and requires banks to take into account the potential magnitude of catastrophic events.

Volatility Clustering and Fat Tails are two of the Key Technical Challenges.

Two empirical features of financial time series that are well known complicate the accurate prediction of risk measures such as VaR and CVaR: volatility clustering and "fat tails" (leptokurtosis). Volatility clustering is the phenomenon that high-volatility events are likely to be followed by additional high-volatility events implying that the volatility of the market is time varying and heteroskedastic. Models that don't take this factor into consideration usually do not adjust fast enough to increase market stress.

Moreover, financial returns are in general not normally distributed. Instead they have fat tails; that is, extreme price moves are much more common than a normal distribution would suggest. This disparity can cause the systematic underestimation of tail risk. With these two attributes, along with regime switching (the underlying model generating process may shift during a crisis), a variety of standard parametric and non-parametric models can be ineffective in predicting. A framework capable of capturing the clustering of volatility, the non-normality of returns, and of capturing structural shifts in the market regime is necessary to be able to manage risk effectively.

Model Gaps: Critiques of Current Approaches. Model Gaps: Critiques of Current Approaches.

Although risk modelling has come a long way, there are still some gaps in risk modelling practice. Although the simplicity and non-parametric approach of Historical Simulation has made it a popular approach, it is a very backward-looking method and makes the assumption that the future will resemble a window of recent history. It tends to lag in times of rapid market changes. On the other hand, parametric GARCH-type models are able to

model volatility clustering very well, but they are very sensitive to the error distribution; if it is mis specified, the parametric VaR and CVaR estimates are biased.

More sophisticated methods such as Filtered Historical Simulation try to do one better by using the GARCH volatility together with empirical remaining , but are usually based on the assumption of one market regime. Until now, there has not been much research comparing the effectiveness of these "regime-aware" Monte Carlo methods, which adapt to the evolving market environment. Moreover, little research has been done that directly compares these approaches in both developed and emerging markets, that can possess vastly different tail behaviours and volatility profiles.

The contributions section is about to begin. The contributions section is now beginning.

To answer these questions, this study compares the performance of the two regime-aware Monte Carlo and Filtered Historical Simulation models with the traditional models. We examine eight selected equity markets around the world, ranging from developed (such as the United States, United Kingdom and Japan) to emerging (such as Brazil, India, China and others).

This paper makes three contributions:

1. Introduction of Regime-Awareness: We add Hidden Markov Models to MC and FHS to enable the models to switch parameters depending on the regime that is detected (e.g., bull vs. bear).
2. Cross-Market Robustness: We explore whether certain model architectures outperform in all markets or if they perform better in the more mature and liquid markets.
3. Rigorous Statistical Validation: We use a complete set of back testing procedures including VaR tests (Kupiec, Christoffersen) and CVaR tests (Acerbi-Szekely), to comply with strict regulatory requirements.

The study connects the dots between regime-switching theory and real-life risk forecasting and offers evidence-based answers to help financial institutions deal with the changing global terrain.

2. Literature Review

This review presents a summary of the development of tail risk modelling, from traditional conditional volatility models to current regime-aware and coherent approaches. The usefulness of a model for regulatory or practical purposes is predicated on how well the model adapts to volatility clustering and structural changes, as discussed in the previous section on the fragile nature of risk management during the Global Financial Crisis and the COVID-19 pandemic.

In this module, students will focus on the concept of value at risk, and on the foundations of historical simulation.

2.1 Value-at-Risk and Foundations of Historical Simulation

Market risk, which has traditionally been defined as the maximum loss on the market over a certain time period with a specific degree of confidence, has been measured most often using Value-at-Risk. There are different approaches to its estimation and one of the most popular is Historical Simulation, which does not require the specification of a functional form for the underlying return distribution (Angelidis et al., 2006). In contrast to simple

Gaussian models (Angelidis et al., 2006), HS accounts for stylized facts, like the fact that the distribution of prices is leptokurtic and skewed.

But the theoretical support for the standard HS is undermined by assuming that returns within the estimation period are independently and identically distributed (i.i.d.). This way, all the historical observations, within a specific time window (typically 250 or 500 days) are assumed to be equally relevant to the present market situation, thus neglecting the conditional aspect of financial risk (Andersen et al., 2013). This non-conditionality creates the “ghost effect” that extreme returns will continue to show up in the rolling sample for the length of the sample window and serve to artificially inflate risk estimates well after the markets have calmed down (Andersen et al., 2013). On the other hand, if a new crisis arises, the model does not have the ability to raise risk buffers quickly because the model is based on a long period of quiet data (Andersen et al., 2013). The inappropriate inclusion of conditional heteroskedasticity into the standard HS is a drawback that renders it poorly suited for the dynamic setting of today's financial markets, because the null and alternative hypotheses the model is evaluating are of unconditional return distribution and current volatility regime, respectively, and these hypotheses are not easily distinguished by standard HS (Pritsker, 2005).

2.2 The GARCH Model Family and Volatility Clustering

As a remedy for some of these shortcomings, the literature has concentrated on modelling the "volatility clustering" observed in empirical data, specifically that large price moves are likely to be followed by additional large moves, and calm periods are likely to last longer (Furman and Wohar, 2019; Engle, 2004; Sahiner, 2022). The ARCH model introduced by Robert Engle is the earliest model in this area, with the conditional variance being a function of past squared residuals (Wang, 2002). In particular, the specification of ARCH(p) yields a high value of current squared returns which boost the market volatility expectations, thus reflecting the volatility persistence of market shocks (Wang, 2002).

On this, Bollerslev (2002) came up with the Generalized Autoregressive Conditional Heteroscedasticity model. The most popular variant is the GARCH specification where forecasts of the variance are a weighted average of three terms: the long run average variance, the variance forecast from the previous period, and the new information supplied by the last shock (Engle, 2004). This mechanism is what makes the model fast to change its state when there is a sudden turbulence and slowly return to the mean in the other periods.

Since the ARCH/GARCH framework proved to be a success, many variants have been devised to model certain market characteristics, as Engle (2004) dubbed the variants an "alphabet soup". Examples are the Exponential GARCH model, which has been developed to incorporate the "leverage effect" (i.e., negative returns affecting volatility more than positive returns) (Engle, 2004). Further extensions of this include the ARCH model of Asymmetric Power ARCH (APARCH) and GJR-GARCH models, which further elaborate the relationship between shocks and conditional variance (Geenens & Dunn, 2020). These models are powerful, but selection of the innovation distribution greatly influences their performance as stated in modern literature. It is also known that the inclusion of skewed Student's t-distribution or Extreme Value Theory (EVT) with GARCH models can lead to better modelling of “fat tails” or leptokurtosis, which are commonly found in equity returns (Kamronnahr et al., 2024; Shayya et al., 2023).

2.3 Filtered Historical Simulation as a Hybrid Approach

Metastatistical Simulation to Visualize Risk and the Estimation of the Upper 10% Value in Exceedance. Metastatistical Simulation as a Mixed Methodology (Barone-Adesi et al., 1999) introduced Filtered Historical Simulation, which recognizes that there are distributional flexibility/compromises between HS and the dynamic responsiveness of GARCH. FHS is a mixed approach between parametric and non-parametric approaches (Marimoutou et al., 2009). It consists of fitting a conditional volatility model (usually a GARCH specification) to the return series to obtain standardized residuals. These residuals are "filtered" from their volatility dynamics and then simulated in the future using bootstrap resampling (Nieto and Ruiz 2016).

The main advantage of FHS is to be able to obtain risk measures that are consistent with the state of the market without forcing the residuals to follow a certain parametric distribution (Marimoutou et al., 2009). Unlike VaR, FHS can provide estimates of the VaR at any given level of confidence, and it can cope with distributional characteristics of empirical data that are not normal (Marimoutou et al., 2009). Empirical evidence has been consistently showing that FHS outperforms standard HS, especially in situations of high volatility and financial crises, where the static approach cannot reflect the very dynamic nature of the risk landscape (Bautista & Mora, 2021). Compared to other studies, FHS has been found to yield the strongest risk estimation in emerging markets due to its ability to reconcile the conditional heteroskedasticity with the heavy-tailedness of the emerging markets returns (Bautista & Mora, 2021).

2.4 Regime-Switching and Hidden Markov Models

The difference between FHS-GARCH models and other FHS models is that the former have a constant underlying process. But, financial markets often experience structural breaks where the underlying process of how prices are formed changes abruptly, such that a single-regime model is not applicable (Ndlovu & Chikobvu, 2024). Hamilton (Hamilton, 1989) initiated the use of regimeswitching models and this concept has been embraced by the literature to overcome this problem. The models assume that the market is characterized by distinct and unobserved latent states (such as bull vs. bear markets) (Diebold & Rudebusch, 1996). The transitions between the states are modeled by a first-order Markov process so that the model can learn the evolution of the mean and the variance (Nasr et al., 2016).

The problem of "spurious volatility persistence" is found if a single state model is extended to deal with data with structural breaks (Cai, 1994). At short horizons, Markov Regime-Switching GARCH models can be seen to outperform standard GARCH, since they better represent the regime-dependency of volatility, as was proven by Marcucci (2005). Moreover, the Hidden Markov Model has been found to be useful at identifying different market regimes (Pallotta & Ciciretti, 2023).

The advantages are offset by the drawbacks of regime-aware models. One of the main objections to many basic HMM specifications is that these specifications only change the return distribution in discrete steps but assume the volatility is constant in one step, as opposed to GARCH models where the conditional volatility is updated every day. Therefore, interactions between regimes are not captured by a model and it can be outperformed by simpler GARCH models (Birtoiu & Dragu, 2012). In recent years a

number of advances have paved the way for the inclusion of these more complicated architectures, namely the integration of Markov-switching dynamics with volatility filtering as in the GARCH family, as achieved in the joint estimation outlined above in the package MSGARCH in R (Ardia et al., 2019).

2.5 Expected Shortfall and Coherent Backtesting

paradigms have changed, as have metrics. In both, the metrics have changed as much as the paradigms. Artzner et al. (1999) articulated the need for "coherent" risk measures, noting that VaR does not meet this criterion, subadditivity, which would imply that the risk of a collective portfolio would not necessarily be sum of the portfolios' risks. In addition, VaR is "tail-blind" in the sense that it does not take account of whether a loss takes place at the tail or not, meaning that it does not reflect the significance of the losses beyond the chosen quantile (Acerbi & Tasche, 2002). Expected loss based on a large number of exceedances over the VaR threshold (EWT) improves on these shortcomings by meeting the requirements of subadditivity and being sensitive to extreme tail events (Acerbi & Tasche, 2002).

The Basel Committee (Assa et al., 2023) introduced the Fundamental Review of the Trading Book in which they made VaR an inferior approach to ES (at the 97.5% confidence level). The introduction of ES provides higher complexity of calculations (Chang et al., 2017) and also some regulatory complexity in the implementation as it enhances the standards expected from the models during stress but weakens them during normal times. The internal models are required to undergo rigorous desktop tests with profit and loss attributions in order to gain regulatory approval (McCullagh et al., 2022). Back testing ES is a more difficult exercise than VaRs because the models for eliciting VaRs and ES are often combined (Owusu et al., 2021). To overcome this, Acerbi and Székely (Acerbi & Székely, 2014) present model independent and non-parametric back-testing procedures based on Monte Carlo simulations, which are by now the common choice to validate ES forecasting methods (Novales & Garcia-Jorcano, 2018). The approach takes into account the experience of comparing actual losses with the foreseen ES level, making the methods "distribution free" (Novales & Garcia-Jorcano, 2018). E-backtesting (Wang et al., 2025) is another modern approach that leverages e-values to validate risk forecasts in a model-free way and scoring function tests can accommodate the estimation errors in a set of joint VaR-ES models (Deng & Qiu, 2021). For example, scoring function tests cater for the estimation errors in the models of joint VaR-ES (Deng & Qiu, 2021) and the e-backtesting approach involves the use of e-values to validate risk forecasts model-free (Wang et al., 2025).

2.6 Productivity and Risk in Emerging Markets

In emerging markets in particular, advanced risk models are crucial, as these tend to be unique. Typically, the volatility of the emerging markets share indices is higher than that of the 'Western' share indices, such as the Bovespa index indicating annualised volatility of 26.0% while the FTSE 100 reveals volatility of 17.7% annually (USER_DOCUMENT). In addition, normality as assumed in standard situations is often not accepted in EMs. Harvey (1995) discovered that the normality hypothesis was not rejected in the 20 developed countries studied but was rejected at the 5% level in 14 of the developing countries considered.

In addition, there is a considerable non-linear dependence and fat tails feature in the emerging market. (Gençay & Selçuk, 2004) Gençay and Selçuk have pointed out that in EMs, distributions of daily returns are in general often different in moment properties at the right and left tails, which means that risk and reward are no longer symmetrically distributed. Although EMs have advantages in terms of diversification, particularly during bear markets, they have been getting firmly entangled in tails over time (Christoffersen et al., 2012).

The approaches traditionally applied to medium-term weather forecast combining have been shown to misperform during situations where medium-term weather conditions are highly instable or possibly due to structural breaks, in the EMs. (Le, 2024) This requires the use of particular instruments like joint loss functions proposed by Owusu et al. (Owusu et al., 2021) to compare the models of elicitable joint VaR and ES. The researchers discovered that there is extreme market-condition dependency and heterogeneity in these risks in China and South Africa, meaning that modeling these tail risks for them will be much more difficult than for other markets (Owusu et al., 2021). In this study, we aimed to see if regime-aware battery or continuously filtering the volatility provides better risk mitigation across these diversified global environments using a unified backtesting battery that integrates the work of Kupiec (1995), Christoffersen (1998), and Diebold-Mariano (1995) tests (Owusu et al., 2021).

3. Data

The adjusted close prices of eight stock market indexes over roughly 20 years were downloaded, and stored in comma-separated value (CSV) snapshots (snapshot dated 2026-06-16). These snapshots are the only ones the research pipeline can decode and they cannot be used to get live data, thus enabling correct reproducibility of the work; per-file SHA-256 hashes, row counts, date ranges are recorded in a dataset manifest. Throughout, daily simple returns are used. A priori, the eight indices are classified into their 4 groups of developed markets (S&P 500, DAX, FTSE 100, Nikkei 225) and emerging markets (Nifty 50, Sensex, Bovespa and JSE Top 40).

Data is not uniformly covered, and is reported transparent. Most series span 2006-06-16 to 2026-06-16; the exceptions are Nifty 50, whose Yahoo Finance history begins 2007-09-17, and JSE Top 40, whose snapshot ends 2026-06-12. The results are presented in descriptive statistics in Table 1. Two characteristics are important for the results that follow: first, all the series experience strong negative tail events, ranging from the worst day return on 2008 and 2020 be 11.6%, to be 14.8% respectively; second, emerging markets have a moderately higher underlying volatility than developed markets, namely 17.7% versus 26.0% for the FTSE 100 and the Bovespa index, respectively. The distributions of daily-return and the baseline volatilities are shown in figure 4 by per market.

Figure 4. Market return distributions and baseline volatility (frozen snapshot 2026-06-16)

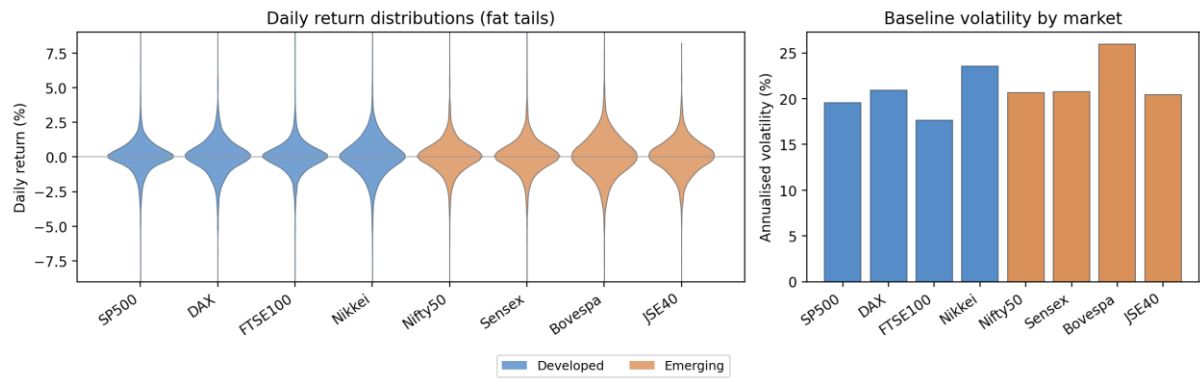


Table 1. Descriptive statistics (daily simple returns, frozen snapshot 2026-06-16).

Market	Symbol	Group	Start	End	Obs	Mean (%)	Std (%)	Ann. Vol (%)	Worst Day (%)
S&P 500	^GSPC	Developed	2006-06-16	2026-06-16	5029	0.043	1.231	19.5	−11.98
DAX	^GDAXI	Developed	2006-06-16	2026-06-16	5071	0.039	1.320	21.0	−12.24
FTSE 100	^FTSE	Developed	2006-06-16	2026-06-16	5049	0.019	1.113	17.7	−10.87
Nikkei 225	^N225	Developed	2006-06-16	2026-06-16	4888	0.043	1.484	23.6	−12.40
Nifty 50	^NSEI	Emerging	2007-09-17	2026-06-16	4595	0.045	1.300	20.6	−12.98
Sensex	^BSESN	Emerging	2006-06-16	2026-06-16	4911	0.050	1.309	20.8	−13.15
Bovespa	^BVSP	Emerging	2006-06-16	2026-06-16	4949	0.046	1.636	26.0	−14.78
JSE Top 40	^J200JO	Emerging	2006-06-15	2026-06-12	4979	0.044	1.289	20.5	−9.92

4. Methodology

Figure 1. Research workflow

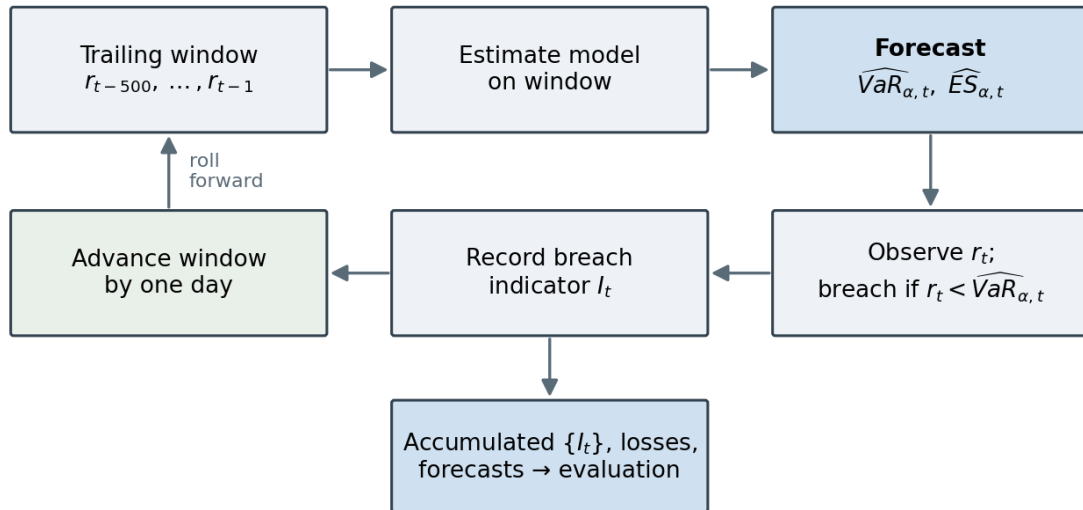


4.1 Forecasting protocol

Let r_t be the daily simple return of an index. Given a certain level of probability $\alpha = \{0.05, 0.01\}$ each of these models generates (signed) one-step-ahead conditional VaR (α, t) and (signed) conditional expected shortfall (α, t). The forecasts are made in a time-series fashion in what is called a rolling walk-forward approach: at the current day t , each model is fitted to the previous $w = 500$ data points ($r_{t-w}, \dots, r_{t-1}$) and is then used to predict r_t , w is then advanced by one day, t is the next day, and the models are re-estimated using the new window of data. No data is used after time t for the forecast for time t , meaning that the evaluation is purely out-of-sample evaluation. If the realised return is less than the forecast VaR, $r_t < \text{VaR}(\alpha, t)$, the indicator $I_t = 1 \{r_t < \text{VaR}(\alpha, t)\}$ will be evaluated and will reflect having had a breach (or exceedance).

4.2 Forecasting models

Figure 2. One-step VaR/CVaR forecasting pipeline (per market, per model)



The following 5 models are compared: They have the identical rolling protocol and only distinct in how the distribution of the next day is estimated.

Gaussian. Estimates of the window mean μ and standard deviation σ are estimated and the return assumed to be normally distributed. The VaR and the expected shortfall are respectively $\text{VaR}_\alpha = \mu + \sigma\Phi^{-1}(\alpha)$ and $\text{ES}_\alpha = \mu - \sigma\phi(\Phi^{-1}(\alpha))/\alpha$, where the Φ^{-1} is the standard normal quantile function and the density function ϕ of the standard normal distribution are denoted. This is a slim-tailed parameter baseline

Historical Simulation. The VaR is the α -quantile of the window returns and the expected shortfall is the mean of the window returns that are $\leq$ VaR. Nothing is assumed about the distribution, the method retains whatever skew and kurtosis the window has, but considers all the observations in the window as exchangeable.

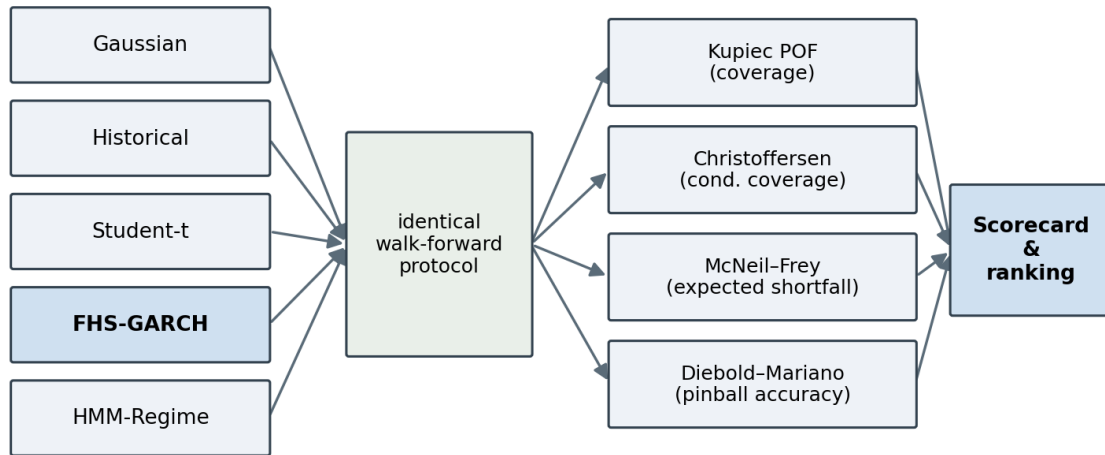
Student-t. The window is fitted to a Student-t distribution with parameters (ν, μ, σ) using maximum likelihood estimation. The window is modeled with a Student-t distribution, with parameters (ν, μ, σ) , where it is fitted using maximum likelihood. The VaR is taken as rather the fitted t α -quantile, while the expression of the shortfall of the Student-t distribution is closed-form and used in the expected shortfall. This model admits heavier-than-normal tails thus deepening the VaR as compared with the Gaussian model on small α .

Filtered Historical Simulation combined with GARCH volatility filter (FHS-GARCH). The window is fit with a constant-mean GARCH(1,1). The conditional variance of the window corresponds to a conditional standard deviation of ε_t of $\sigma_t^2 = \omega + a\varepsilon_{t-1}^2 + b\sigma_{t-1}^2$. The standardised residuals $\hat{z}_t = (r_t - \mu)/\sigma_t$ are created. These residuals are closer to independent and identically distributed (i.i.d.) than are the raw returns given that the volatility dynamics have been removed. The VaR is simply the empirical α -quantile of the standardised residuals scaled by the volatility forecast at time t , $\hat{\sigma}_{t+1}$: $\text{VaR}_\alpha = \mu + \hat{\sigma}_{t+1} Q_\alpha(\hat{z})$; where Q_α is the residual quantile and the volatility forecast, $\hat{\sigma}_{t+1}$, rescales the mean of the tail of the residuals in the same manner as the VaR. Combines a non-parametric tail shape with a volatility forecast that is dynamic and is continuously updated.

Hidden Markov model- regime mixture (HMM-Regime). The window is fitted by a K -state gaussian HMM (μ_k, σ_k^2, A) . The state-probability vector at the end of the window (γ^T) is propagated by the transition matrix for one time-step, in order to calculate the next-day state distribution vector $\xi = \gamma^T A$. The next day density is $\sum \xi_k N(\mu_k, \sigma_k^2)$, the VaR and the expected short and are taken from the α -tail of this mixture. This is the Monte Carlo model under test, the volatility being assumed to be equally distributed between different regimes but variable between the regimes.

4.3 Evaluation

Figure 3. Model-comparison framework: five engines, one protocol, four tests



A VaR forecast is a falsifiable claim, so each model is evaluated by four formal tests rather than by a single summary number.

Kupiec proportion-of-failures (POF) test. The sequence of indicators for the breach is Bernoulli, with probability of success α , under a properly specified model. Under the same assumptions, the unconditional-coverage likelihood-ratio statistic is asymptotically $\chi^2(1)$: $LR_{uc} = -2 \cdot \ln[(1-\alpha)(N-x)\alpha x / (1-\hat{\pi})(N-x)\hat{\pi}x]$ where $\hat{\pi} = x / N$. It checks if the rate of the defect is α .

Christoffersen conditional-coverage test. Just coverage is not enough: a model may fail at the right speed, but its failures may happen in crises. Based on the numbers of breaches in the preceding day's transition sequence, the Christoffersen independence test is a likelihood ratio test statistic $LR_{ind} \sim \chi^2(1)$ that could determine whether there is any "dependence" today in the presence of a breach yesterday. In this study, the joint conditional-coverage statistic $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$ tests for both correctness and independence, while the former is the more significant, in this case, because it is more relevant to the data.

McNeil-Frey expected-shortfall test. The coverage tests only test the VaR quantile and do not provide any information regarding the expected shortfall. The McNeil-Frey procedure compares exceedance residuals on breach days, $r_t - ES(\alpha, t)$, that should have a mean of zero if ES is properly calibrated. A one sided bootstrap test of the null hypothesis of zero mean is applied and a significantly negative mean, suggesting that realised tail losses are systematically more than the forecast shortfall.

Diebold-Mariano (DM) predictive-accuracy test. The difference in the per-day loss between two models is estimated using the pinball (quantile) loss $L_\alpha(r, q) = (\alpha - 1\{r < q\}) \cdot (r - q)$, with the per-day loss estimated for the VaR forecast under each model, and is tested by the DM statistic, which accounts for the long-run variance, with the Newey-West correction and the Harvey-Leybourne-Newbold small-sample correction. The test is conducted twice at the 5% significance level and each model pair judges a winner or a tie in each market.

5. Results

The calibration scorecard, which combines all eight markets, is presented in Table 2; the per-market conditional-coverage p-values are displayed as a heatmap in Figure 5; and predictive-accuracy wins are summarised in Figure 6.

Table 2. Calibration scorecard for eight markets.

The cells with CC = markets passing Christoffersen conditional coverage ($p > 0.05$), the cells with ES = cells passing the McNeil–Frey test out of 16 cells ($8 \text{ markets} \times 2 \text{ levels}$), the cells with DM wins = significant pairwise Diebold–Mariano pinball-loss wins at $\alpha = 5\%$.

Model	CC 5% dev	CC 5% emg	CC 1% dev	CC 1% emg	ES /16	DM wins
Gaussian	0/4	1/4	0/4	0/4	0/16	3
Historical	0/4	1/4	0/4	0/4	13/16	9
Student-t	0/4	1/4	0/4	1/4	14/16	3
FHS-GARCH	4/4	4/4	0/4	4/4	16/16	32
HMM-Regime	0/4	0/4	0/4	2/4	15/16	7

5.1 Calibration at the 5% level.

FHS-GARCH does pass at 5% in all markets (8 of 8), or in the case of being on the border, a much higher conditional coverage value when used in every market. The Gaussian, Historical and Student-t models pass only 1 emerging market while the HMM-Regime model passes none. The reason why the four competing models fail is mainly due to their failure of independence rather than to their failure of rate – the unconditional moments of failure rates of these four models are almost always close to the nominal 5% value (for the S&P 500, both the Gaussian and the Historical model fail at 5.3%, the latter passes the Kupiec test with $p > 0.39$, while the latter fails at 5.2%), and their breaches are concentrated in the high-volatility periods so the Christoffersen independence component is rejected (Christensen's independence $p \approx 0$). FHS-GARCH is the only unconditional filter which allows the largest breaches in the short term during periods of high volatility, without the need to tune it.

5.2 Calibration at the 1% level.

The deeper tail is of greater interest. Conditional coverage in all 4 emerging markets (4/8) and in both not in FHS-GARCH in all 4 developed markets (4/8). There are no other models with measure at two or more of the eight measures at 1%, so the models on this chart are not statistically significant. The HMM-Regime model underperforms at the 5% level but achieves its best result of two clear passes at 1%, in the emerging markets.

Figure 5. Christoffersen conditional-coverage p-values by model and market

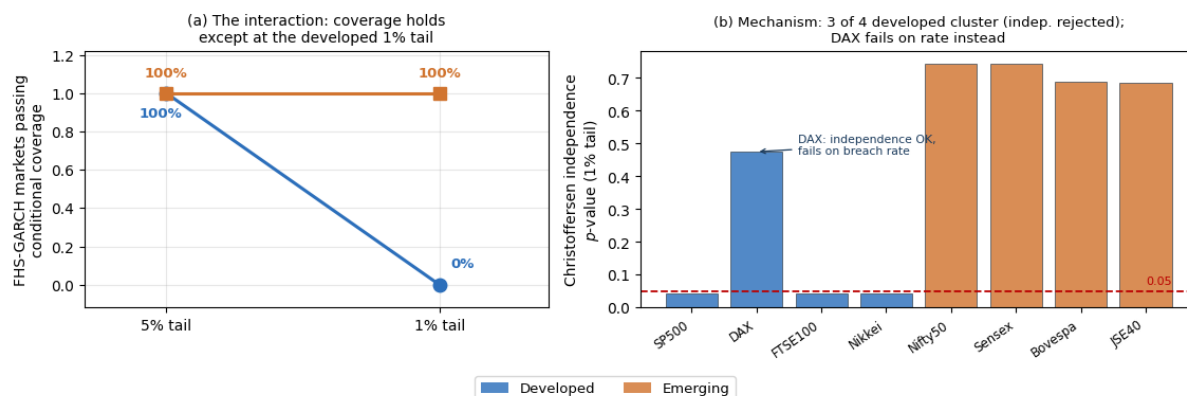
Conditional-coverage p-value ($\alpha = 5\%$); green = pass ($p > 0.05$); line separates developed emerging								
Gaussian	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.07
Historical	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.13
Student-t	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.10
FHS-GARCH	0.69	0.10	0.08	0.38	0.19	0.47	0.94	0.13
HMM-Regime	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	S&P500	DAX	FTSE100	Nikkei	Nifty50	Sensex	Bovespa	JSE40

Conditional-coverage p-value ($\alpha = 1\%$); green = pass ($p > 0.05$); line separates developed emerging								
Gaussian	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Historical	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.02
Student-t	0.00	0.00	0.00	0.00	0.16	0.00	0.00	0.00
FHS-GARCH	0.03	0.00	0.03	0.02	0.14	0.21	0.35	0.38
HMM-Regime	0.00	0.00	0.00	0.01	0.06	0.15	0.03	0.05
	S&P500	DAX	FTSE100	Nikkei	Nifty50	Sensex	Bovespa	JSE40

5.3 The developed-market 1% failure.

The Christoffersen test is used in this study to decompose the failures of the FHS-GARCH model and to isolate the mechanism in the model, which is non-uniform. In three of the four developed markets (S&P 500, FTSE 100, Nikkei 225) the breach rate is close to nominal (for the S&P 500, 58 breaches in 4,529 observations; Kupiec $p \approx 0.07$, passing) but the breaches are serially dependent (independence $p \approx 0.044$, failing): they arrive consecutively within crisis episodes. The GARCH filter increases the VaR in case of a shock, but even after a crash gaps down, it is not able to increase the VaR quickly enough to avoid successive breaches of the threshold as shown in the per-market breach plots in Figure 8. There is only one exception: DAX attains it by lack of clustering but has a too high (unconditional) breach rate (1,60% for 0,48% of independence) instead. The advantage of both is or is not a covered breach, but perhaps the most common developed market configuration is breach clustering: tail-depth $\times$ market-type, as evinced in Figure 7. The emerging markets, which consistently have higher baseline volatility, do not cluster nor do they over-break at the 1% level, remaining (at all times) calibrated.

Figure 7. Tail-depth $\times$ market-type interaction for FHS-GARCH (frozen snapshot 2026-06-16)

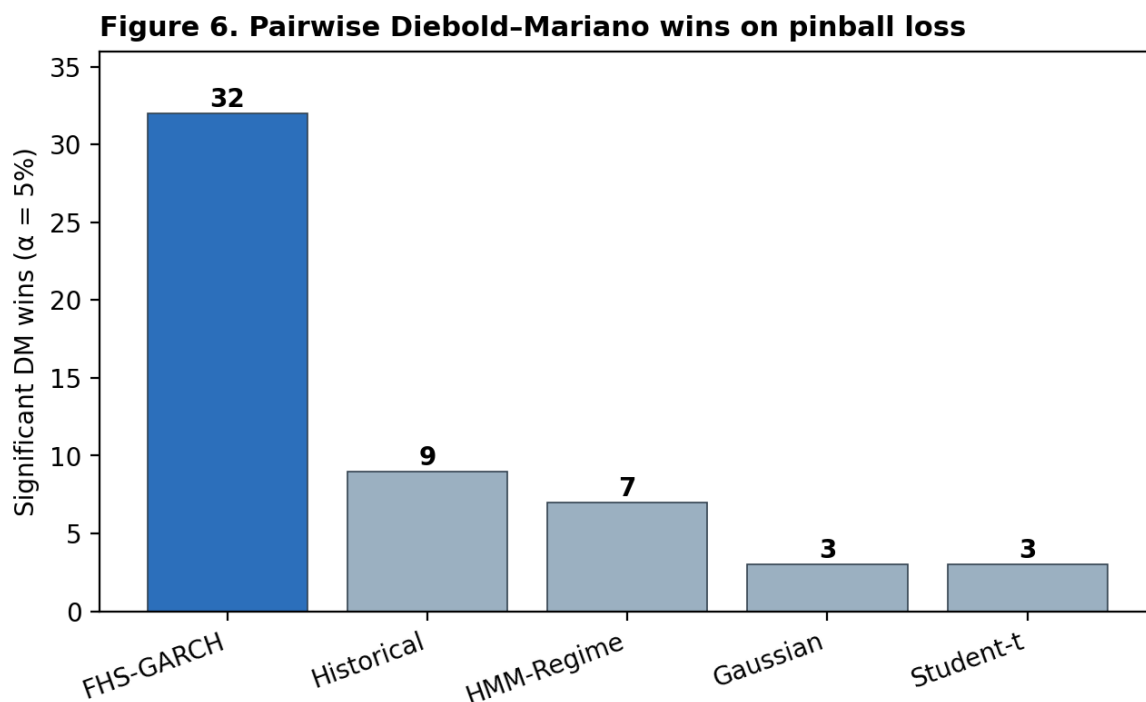


5.4 Expected Shortfall.

All 3 models SBP-FHS-GARCH, Student-t, and HMM-Regime satisfy the 50% expected-shortfall test in their respective cells. In all 16 cells, FHS-GARCH passes the tests for McNeil–Frey's expected-shortfall, while both Student-t and HMM-Regime pass in 14 and 15 cells, respectively. In fact, the Gaussian model fails to do all 16; the standard model has very thin tails, so that the deficit is systematically underestimated, precisely the thing for which the expected-shortfall backtest is aimed. Expected-shortfall results therefore follow but do not just repeat the coverage results, but also flag the tail size miss of the Gaussian model, not its coverage miss clustering.

5.5 Predictive Accuracy.

FHS-GARCH holds significant superiority to Historical in 32 of its 41 pairwise comparisons whereas it is only marginally superior to HMM-Regime, Gaussian, and Student-t in 7, 3, and 3, respectively, when we draw the Diebold–Mariano comparisons at the 5% level (Figure 6). Calibration (coverage and shortfall) and predictive accuracy (pinball loss) are two distinct evaluation philosophies and their joint performance for FHS-GARCH is considered to be in agreement and a very good corroboration: shortfall calibrated models simultaneously perform best and pinball loss based models perform best; and regime-aware models dominate on both the calibration and the predictive accuracy.



6. Discussion

The research question is answered by saying "No". Put simply, an explicit regime identification (as defined by a Gaussian HMM) worsens a GARCH volatility filter and is the poorest model on 5% conditional coverage across the eight considered equity markets. It seems that the modelling of conditional volatility is consistently more defensible than the use of discrete regimes having fixed within-resolution volatility parameters. While the HMM resets the return distribution only in "discrete jumps" it keeps volatility constant within a regime, the FHS-GARCH updates the conditional volatility on a daily basis; this

is where volatility clustering is prevalent in the markets, and this is where calibration shines.

The following interpretation would be relatively limited. Because the HMM used here does not have in-regime volatility dynamics, there is no result found of the fact that regime information is intrinsically valueless, but only of the fact that a volatility filter does not add anything with regard to this regime. The model incorporates both mechanisms in one, that of Markov switching GARCH which incorporates a GARCH process in each regime; the rationale of this study is that regime identification is not necessary on top of a volatility filter – and it is the next natural experiment (Section 8).

The secondary finding includes a real-life relevant warning. Even the highest (1%) caliper of the best method fails in developed markets, mostly due to clustering of these breaches during the crisis period (and, in one developed market, due to a higher breaches count). A well-calibrated 5% VaR system in all eight markets cannot be taken for granted to be well-calibrated at the 1% level where extreme risk is most likely to have an impact — behaviour of extremely exceeded VaR events is to be expected to be poorly behaved, owing to the lack of heavy tails.

7. Limitations

The conclusions drawn from this study are subject to various limitations. It uses a single, 20-year, window. The developed-market results are partly a reflection by the two crises falling within that period, in 2008 and 2020. An analysis of rolling-origin windows would strengthen the robustness of the findings. The modest number of market failures and small to clustered number of exceedances also limit the conclusions drawn. A further caveat concerns independence failures, which is based on a small, clustered number of exceedances.

The design of simultaneous hypothesis tests, involving eight markets, five models and two coverage levels necessarily inflates the family-wise error rate. The most strong claims, therefore, concern extremes: the FHS-GARCH passes eight of eight at 5% level, while none passed the HMM-Regime test. Isolated single-market passes should be read cautiously and ideally subject to a multiplicity correction. When reporting, the HMM specification and construction is handicapped by its lack within-regime volatility dynamics.

8. Future Work

The most direct extension of MS-GARCH models is the Markov-switching GARCH models in which a latent regime variable switches between a set of regimes, each with its own GARCH process. A comparison of the MS-GARCH and FHS-GARCH that aims to isolate the effects of regime detection once conditional volatility is already modelled is worthy of investigation. This paper poses this question but does not have the resources to answer it. Sub-period robustness analyses of 2008 and 2020, perhaps including crises, some multiplicity-corrected confirmatory testing and deepest-tail analyses using longer histories or pooled cross-market exceedances at the 1%-level are possible future avenues for deeper robustness and validity.

9. Conclusion

This paper asks if regime-aware Monte Carlo models can offer better Value-at-Risk (VaR) and expected shortfall (ES) forecasts than both classical and volatility-filtered alternatives, within the same out-of-sample protocol and over four different evaluation tests,

using data from eight developed- and emerging-market equity markets spanning nearly two decades. The answer is negative and consistent across evaluation philosophies: GARCH-filtered historical simulation is the only model to achieve correct conditional coverage at the 5% level in all eight markets, as well as the only model with fully calibrated expected shortfall and the clear winner on predictive accuracy, while the hidden-Markov regime model is the weakest on 5% conditional coverage. A regime mixture with static within-regime volatility therefore offers no calibration advantage over a continuously updating volatility filter. A secondary and practically important finding is that even the best model loses 1% tail calibration in developed markets, where rare exceedances cluster within the 2008 and 2020 crises, while remaining calibrated in higher-volatility emerging markets. Taken together, the evidence shows that, for tail-risk forecasting in broad equity indices, modelling conditional volatility dynamics matters more than identifying discrete regimes - and that the marginal value of regime identification on top of a volatility filter, via Markov-switching GARCH, is the question most worth pursuing next.

Author Contributions

This work was done by Vishwa Jadhav. Vishwa Jadhav is the sole author. The author initiated and directed the study, made all methodological and design decisions, executed and independently verified every experiment against the frozen dataset and prepared the manuscript.

Reference

- Acerbi, C., & Székely, B. (2014). *Backtesting expected shortfall: Introducing three model-independent, non-parametric back-test methodologies for expected shortfall*. MSCI Research Insight. https://www.msci.com/resources/research/articles/2014/Research_Insight_Backtesting_Expected_Shortfall_December_2014.pdf
- Acerbi, C., & Tasche, D. (2002). On the coherence of expected shortfall. [https://doi.org/10.1016/s0378-4266\(02\)00283-2](https://doi.org/10.1016/s0378-4266(02)00283-2)
- Andersen, T. G., Bollerslev, T., Christoffersen, P., & Diebold, F. X. (2013). Financial risk measurement for financial risk management. In *Handbook of the Economics of Finance* (pp. 1127–1220). <https://doi.org/10.1016/b978-0-44-459406-8> [verify]
- Angelidis, T., Benos, A., & Degiannakis, S. (2006). A robust VaR model under different time periods and weighting schemes. *Review of Quantitative Finance and Accounting*, 28(2), 187–201. <https://doi.org/10.1007/s11156-006-0010-y>
- Ardia, D., Bluteau, K., Boudt, K., Catania, L., & Trottier, D.-A. (2019). Markov-switching GARCH models in R: The MSGARCH package. *Journal of Statistical Software*, 91(4). <https://doi.org/10.18637/jss.v091.i04>
- Artzner, P., Delbaen, F., Eber, J., & Heath, D. (1999). *Coherent measures of risk*. <https://doi.org/10.1111/1467-9965.00068>
- Assa, H., Lin, L., & Wang, R. (2023). Calibrating distribution models from PELVE. *North American Actuarial Journal*, 28(2), 373–406. <https://doi.org/10.1080/10920277.2023.2211649> [verify]

- Barone-Adesi, G., Giannopoulos, K., & Vosper, L. (1999). VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets*, 19(5), 583–602. [DOI verify]
- Bautista, R. S., & Mora, J. A. N. (2021). Value-at-risk predictive performance: a comparison between the CaViaR and GARCH models for the MILA and ASEAN-5 stock markets. *Journal of Economics, Finance and Administrative Science*, 26(52), 197–221. <https://doi.org/10.1108/jefas-03-2021-0009> [verify]
- Birtoiu, A., & Dragu, F. G. (2012). *The performance of time-varying volatility and regime switching models in estimating Value-at-Risk*. Lund University Publications Student Papers (Lund University). <http://lup.lub.lu.se/student-papers/>
- Cai, J. (1994). A Markov model of switching-regime ARCH. *Journal of Business and Economic Statistics*, 12(3), 309–316. <https://doi.org/10.1080/07350015.1994.10524545> [verify]
- Chang, C., Jiménez-Martín, J., Maasoumi, E., McAleer, M., & Pérez-Amaral, T. (2017). Choosing expected shortfall over VaR in Basel III using stochastic dominance. *Advances in Economics, Business and Management Research*, 26, 95–113. [DOI verify]
- Christoffersen, P. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862. <https://doi.org/10.2307/2527341> (source showed 841 only)
- Christoffersen, P., Errunza, V. R., Jacobs, K., & Langlois, H. (2012). Is the potential for international diversification disappearing? A dynamic copula approach. *Review of Financial Studies*, 25(12), 3711–3751. [DOI verify]
- Danielsson, J., & de Vries, C. (2000). Value-at-risk and extreme returns. *Annales d'Économie et de Statistique*, 60, 239–270. <https://doi.org/10.2307/20076262> (source showed 239 only)
- Deng, K., & Qiu, J. (2021). Backtesting expected shortfall and beyond. *Quantitative Finance*, 21(7), 1109–1125. <https://doi.org/10.1080/14697688.2020.1834120> [verify]
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business and Economic Statistics*, 13(3), 253–263. [DOI verify]
- Diebold, F. X., & Rudebusch, G. D. (1996). Measuring business cycles: A modern perspective. *The Review of Economics and Statistics*, 78(1), 67–77. (source showed 67 only; verify)
- Engle, R. (2004). Risk and volatility: Econometric models and financial practice. *American Economic Review*, 94(3), 405–420. <https://doi.org/10.1257/0002828041464597> [verify]
- Geenens, G., & Dunn, R. S. (2020). A nonparametric copula approach to conditional value-at-risk. *Econometrics and Statistics* (Vol. 21, pp. 19–37). <https://doi.org/10.1016/j.ecosta.2020.07.001> [verify]

- Gençay, R., & Selçuk, F. (2004). Extreme value theory and value-at-risk: Relative performance in emerging markets. *International Journal of Forecasting*, 20(2), 287–303. <https://doi.org/10.1016/j.ijforecast.2003.09.005> [verify]
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384. <https://doi.org/10.2307/1912559> (source showed 357 only)
- Harvey, C. R. (1995). Predictable risk and returns in emerging markets. *Review of Financial Studies*, 8(3), 773–816. <https://doi.org/10.1093/rfs/8.3.773>
- Kamronnahr, K., Bellucco, A., Huang, W. K., & Gallagher, C. (2024). *Estimating value at risk and expected shortfall: A brief review and some new developments*. arXiv. <https://doi.org/10.48550/arxiv.2405.06708> [verify]
- Kupiec, P. (1995). Techniques for verifying the accuracy of risk measurement models. *The Journal of Derivatives*, 3(2), 73–84. <https://doi.org/10.3905/jod.1995.407942>
- Le, T. H. (2024). Forecasting value-at-risk and expected shortfall in emerging market: does forecast combination help? *The Journal of Risk Finance*, 25(1), 160–177. <https://doi.org/10.1108/jrf-06-2023-0137> [verify]
- Marcucci, J. (2005). Forecasting stock market volatility with regime-switching GARCH models. *Studies in Nonlinear Dynamics and Econometrics*, 9(4). <https://doi.org/10.2202/1558-3708.1145>
- Marimoutou, V., Raggad, B., & Trabelsi, A. (2009). Extreme value theory and value at risk: Application to oil market. *Energy Economics*, 31(4), 519–530. <https://doi.org/10.1016/j.eneco.2009.02.005> [verify]
- McAleer, M., & Medeiros, M. C. (2008). A multiple regime smooth transition heterogeneous autoregressive model for long memory and asymmetries. *Journal of Econometrics*, 147(1), 104–119. <https://doi.org/10.1016/j.jeconom.2008.09.032> [verify]
- McCullagh, O., Cummins, M., & Killian, S. (2022). Decoupling VaR and regulatory capital: an examination of practitioners' experience of market risk regulation. *Journal of Banking Regulation*, 24(3), 321–336. [DOI verify]
- McNeil, A. J., & Frey, R. (2000). *Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach*. [https://doi.org/10.1016/S0927-5398\(00\)00012-8](https://doi.org/10.1016/S0927-5398(00)00012-8)
- Michaelides, M., & Poudyal, N. (2023). Good risk measures, bad statistical assumptions, ugly risk forecasts. *Financial Review*, 59(2), 519–543. [DOI verify]
- Nasr, A. B., Lux, T., Ajmi, A. N., & Gupta, R. (2016). Forecasting the volatility of the Dow Jones Islamic Stock Market Index: Long memory vs. regime switching. *International Review of Economics & Finance*, 45, 559–571. [DOI verify]
- Ndlovu, T., & Chikobvu, D. (2024). Estimating extreme value at risk using Bayesian Markov regime switching GARCH–EVT family models. In *IntechOpen eBooks*. <https://doi.org/10.5772/intechopen.1004124>

Nieto, M. R., & Ruiz, E. (2016). Frontiers in VaR forecasting and backtesting. *International Journal of Forecasting*, 32(2), 475–501. <https://doi.org/10.1016/j.ijforecast> [verify]

Novales, A., & García-Jorcano, L. (2018). *Backtesting extreme value theory models of expected shortfall*. EPrints Complutense, Universidad Complutense de Madrid, 19(5), 799–825. [DOI verify]

Owusu, P., Alagidede, P., & Tiwari, A. K. (2021). On the elicibility and risk model comparison of emerging markets equities. *Mathematical and Computational Applications*, 26(3), 63. <https://doi.org/10.3390/mca26030063> (verify pages)

Pallotta, A., & Ciciretti, V. (2023). Should you use GARCH models for forecasting volatility? A comparison to GRU neural networks. *Studies in Nonlinear Dynamics and Econometrics*, 28(5), 725–738. <https://doi.org/10.1515/snde-2022-0025> [verify]